MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Initial Evaluation of the Adult Trauma Patient with Open Extremity Fractures

Orthopedic consult within 30 minutes of arrival time.
Page Ortho resident with injury and trauma class

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Dressing should be 0.9NS soaked gauze & wrapped with Ace wrap wrap only

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Do not open extremity fracture wound site more than 2 times

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All patients will be evaluated for updated Tetanus prophylaxis.

Initial Antibiotic Algorithm (Administer first dose < 60 minutes from presentation)

FIRST DOSE WITHIN 60 MINUTES OF PRESENTATION

Type I: Clean wound < 1 cm diameter, simple fracture pattern, no skin crushing Type II: Wound >1 cm without significant soft tissue crushing, no flaps, degloving, or contusion. Fracture pattern may be more complex.	Preferred: Cefazolin 2g (3g if >120 kg) IV Q8H* Cephalosporin allergy or Severe non-IgE Penicillin Allergy#: Vancomycin 15 mg/kg**
Type III: Open segmental fracture or single fracture with extensive soft tissue injury. Also includes injuries >8 hours old. Type III are subdivided into three types.	Preferred: Ceftriaxone 2g IV Q24 Cephalosporin allergy or Severe non-IgE Penicillin Allergy#: Vancomycin 15 mg/kg** + Aztreonam 2g IV Q8H*

DURATION:
24 hours from time of OR if immediate OR

24 hours from time of injury if no immediate OR

FOOTNOTES:
*Requires dose adjustment in renal impairment

** Initial interval based on creatinine clearance: Q8H if CrCl >90, Q12H if CrCl 50- 90, Q24 if CrCl < 50

#: Cefazolin and Ceftriaxone are safe to use in patients with anaphylaxis to penicillins. Cephalosporins should be avoided in history of severe non-IgE mediated reaction (i.e., SJS, TENS) to any beta-lactam. If allergy history unable to be confirmed, alternative regimen may be considered until further history is obtained. Consult Beta-Lactam Allergy Service for assistance/guidance and see Beta-Lactam Allergy Guidance on Stewardship Website.

Additional Considerations for Antibiotic Spectrum:

Extensive Soil Contamination: May consider addition of Metronidazole 500mg IV Q12H

Extensive Water Exposure: May consider switch to Piperacillin/Tazobactam 4.5g IV Q6H*

Known MRSA Colonization: Add Vancomycin IV 15 mg/kg**

Subsequent Antibiotic Prophylaxis Guidelines Prophylaxis Timing and Duration:

1. Additional **Preoperative** IV antibiotic dose should be given 15-60 prior to incision for every surgery
    a. Preferred: Cefazolin 2g IV (3g if ≥ 120 kg)
    b. Cephalosporin allergy or severe non-IgE penicillin allergy: Vancomycin 1 g if < 80 kg, 1.5 g if ≥ 80 kg
2. Antisepsis with 0.7% iodine povacrylex in 74% isopropyl alcohol preferred (doi: 10.1056/NEJMoa2307679). If not available, or if patient allergy, use 2% chlorhexidine gluconate in 70% isopropyl alcohol
3. Continue algorithm antibiotics for 24 hours after **index debridement.** For patients with chronic open wounds awaiting definitive soft tissue wound closure, empiric prophylactic antibiotic therapy is not indicated. If there is concern for infection, ID consultation is highly recommended.
4. Continue antibiotics for 24 hours after **definitive wound closure.** Antibiotics should only be continued past this point if there is concern for ongoing infection, in which case ID consultation is highly recommended.

Open fracture type	Characteristics of Gustilo Type Open Fracture
Type I	Clean wound smaller than 1 cm in diameter, simple fracture pattern, no skin crushing.
Type II	A laceration larger than 1 cm but without significant soft tissue crushing, including no flaps, degloving, or contusion. Fracture pattern may be more complex.
Type III	An open segmental fracture or a single fracture with extensive soft tissue injury. Also included are injuries older than 8 hours. Type III injuries are subdivided into three types:
Type III A	Adequate soft tissue coverage of the fracture despite high energy trauma or extensive laceration or skin flaps.
Type III B	Inadequate soft tissue coverage with periosteal stripping. Soft tissue reconstruction is necessary.
Type III C	Any open fracture that is associated with an arterial injury that requires repair.

Last revised: 11/2024	**Antimicrobial Stewardship Subcommittee:** 10/2024	**P&T:** 11/2024

References

	Citation
1	Buckman SA, et al. Surgical Infection Society Guidelines: 2022 Updated Guidelines for Antibiotic Use in Open Extremity Fractures. Surg Inf 2022;23(9):817-828. DOI: 10.1089/sur.2022.206
2	Whiting PS, et al. Open Fractures: Evidence-based best practices. OTA Int 2024 May;7(3):e313.
3	Rodriguez L, Jung HS, Goulet JA, Cicalo A, Machado-Aranda DA, Napolitano LM. Evidence-based protocol for prophylactic antibiotics in open fractures: improved antibiotic stewardship with no increase in infection rates. J Trauma Acute Care Surg. 2014 Sep;77(3):400-7; discussion 407-8; quiz 524. doi: 10.1097/TA.0000000000000398. PMID: 25159242.
4	Atwan, Yousif MDa,*; Miclau, Theodore MDb; Schemitsch, Emil H. MD, FRCSCa; Teague, David MDc. Antibiotic utilization in open fractures. OTA International 3(1):p e071, March 2020. | DOI: 10.1097/OI9.0000000000000071

Revision History

Date	Revision
12/2021	Revised type and treatment recommendations

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source